

Submission D solution to Problem 3

Problem Statement

Let $w_1, \dots, w_m$ be non-negative real numbers such that $\sum_{i=1}^m w_i = 1$, and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables. For which values of probability p is it true that

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p?$$

Partial Solution

Note: The following represents a partial solution to the problem. As noted at the end of the proof, sufficiency is fully established, but there remains a structural gap in proving the necessity for non-reciprocal values of p .

Theorem 1. *Let $w_1, \dots, w_m$ be non-negative real numbers such that $\sum_{i=1}^m w_i = 1$, and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables. The inequality*

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p$$

holds for all valid weight distributions if and only if $p = 1/k$ for some integer $k \geq 1$ (or $p = 0$).

Proof. For $p = 0$, the condition reduces to $\Pr[0 \geq 0] \geq 0$, which evaluates to $1 \geq 0$ and is trivially true.

Assume $p = 1/k$ for some integer $k \geq 1$. We construct k coupled sequences of random variables $v_i^{(j)}$ for indices $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, k\}$ as follows: For each i , select an index J_i uniformly at random from the set

$\{1, \dots, k\}$, such that the choices $J_1, \dots, J_m$ are mutually independent. Set $v_i^{(J_i)} = 1$, and set $v_i^{(j)} = 0$ for all $j \neq J_i$.

By this construction, for any fixed sequence index j , the variables $v_1^{(j)}, \dots, v_m^{(j)}$ are independent Bernoulli($1/k$) random variables. Define $X^{(j)} = \sum_{i=1}^m w_i v_i^{(j)}$. For each j , the random variable $X^{(j)}$ has the exact same distribution as the original sum $\sum_{i=1}^m w_i v_i$.

Summing across all k sequences yields:

$$\sum_{j=1}^k X^{(j)} = \sum_{j=1}^k \sum_{i=1}^m w_i v_i^{(j)} = \sum_{i=1}^m w_i \sum_{j=1}^k v_i^{(j)}$$

Since exactly one $v_i^{(j)} = 1$ for each i (specifically when $j = J_i$), the inner sum $\sum_{j=1}^k v_i^{(j)}$ equals 1 deterministically. Thus,

$$\sum_{j=1}^k X^{(j)} = \sum_{i=1}^m w_i (1) = 1$$

Because the k identically distributed random variables $X^{(j)}$ sum to exactly 1, at least one of them must evaluate to $1/k$ or greater. Let A_j be the event that $X^{(j)} \geq 1/k$. We have $\Pr \left[\bigcup_{j=1}^k A_j \right] = 1$. By the union bound:

$$1 = \Pr \left[\bigcup_{j=1}^k A_j \right] \leq \sum_{j=1}^k \Pr[A_j] = k \Pr \left[X^{(1)} \geq \frac{1}{k} \right]$$

Dividing by k gives:

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq \frac{1}{k} \right] \geq \frac{1}{k}$$

This establishes sufficiency.

For values of $p \notin \{0\} \cup \{1/k \mid k \in \mathbb{Z}^+\}$, the inequality does not universally hold.

[Gap: The explicit construction of pathological weight distributions providing counterexamples for all non-reciprocal values of p is structurally omitted.] □